

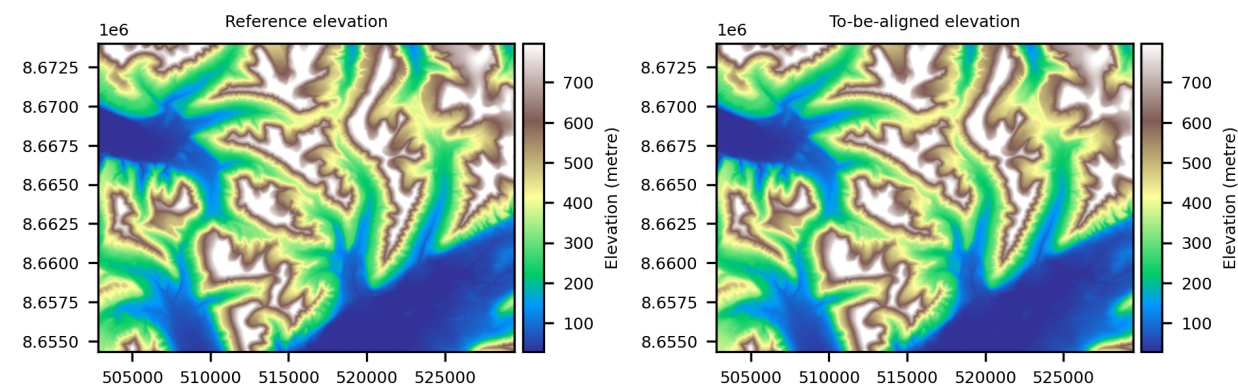
Accuracy assessment report — xDEM

xDEM version: 0.2.3.dev26

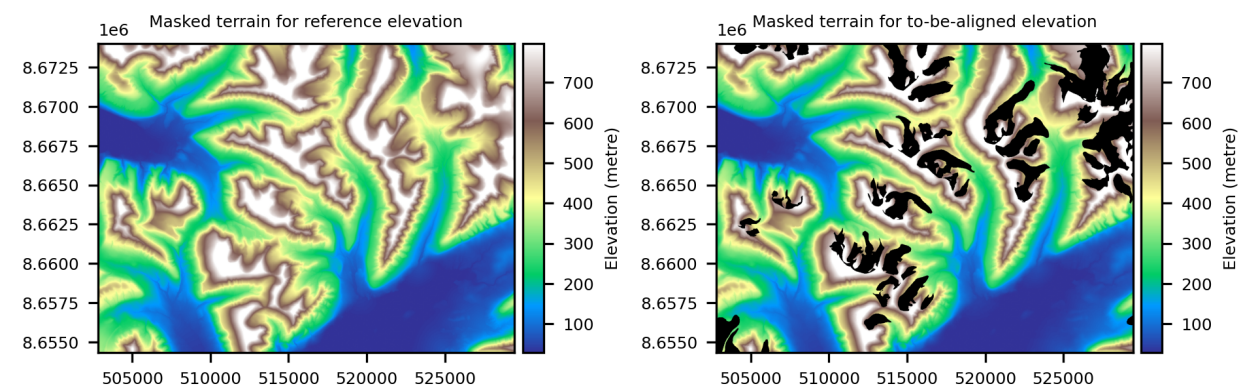
Date: 03/06/2026 11:49:50

Computing time: 12.09 seconds

Elevation inputs



Masked elevation data



Information about inputs

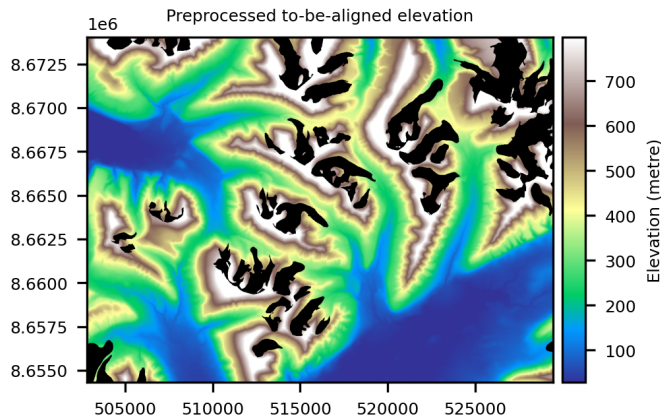
Information	Value
reference_elev	{'path_to_elev': '../xdem/example_data/Longyearbyen/data/DEM_2009_ref.tif', 'from_vcrs': 'None', 'to_vcrs': 'None', 'downsample': '1'}
to_be_aligned_elev	{'path_to_elev': '../xdem/example_data/Longyearbyen/data/DEM_1990.tif', 'path_to_mask': '../xdem/example_data/

```

Longyearbyen/data/glacier_mask/CryoClim_GAO_SJ_1990.shp',
'from_vcrs': 'None', 'to_vcrs': 'None', 'downsample': '1'}
sampling_grid      reference_elev

```

Preprocessed elevation data



Coregistration user configuration

Information Value

```

step_one      {'method': 'LZD', 'extra_information': '{"subsample': 10000}"}
process       True

```

LZD inputs

Information Value

```

random        {'subsample': '10000', 'random_state': '42'}
fitorbin      {'fit_minimizer': 'least_squares', 'fit_loss_func': 'linear'}
iterative     {'max_iterations': '200', 'tolerance': '0.01'}
specific      {}
affine        {'only_translation': 'False'}

```

LZD outputs

Information Value

```

affine        {'centroid': '(515650.83, 8663888.17, 353.11)', 'matrix':
               '[[ 9.99999985e-01 1.17578419e-04 -1.30099078e-04
1.02548967e+01]\n [-1.17594385e-04 9.99999986e-01 -1.22719549e-04

```

```

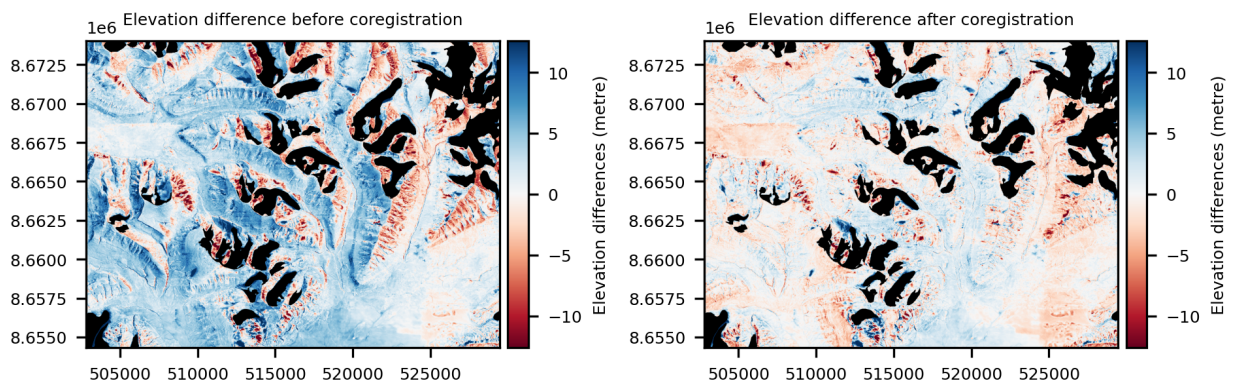
1.90590312e+00]\n [ 1.30084647e-04 1.22734846e-04 9.99999984e-01
-1.93544013e+00]\n [ 0.00000000e+00 0.00000000e+00
0.00000000e+00 1.00000000e+00]]', 'shift_x': '10.25', 'shift_y': '1.91',
'shift_z': '-1.94'}
random      {'subsample_final': '10000'}

```

Statistics

Data	Aligned elevation	Difference before coreg	Difference after coreg
Median	316.330	2.334	-0.048
NMAD	277.103	3.417	2.050
Valid count	1133607	1145221	1133607

Elevation differences



Differences histogram

